targets, detection of concealed targets and so on.

Impact of smarter, sensor-enabled robots in defence

Smarter, sensor-enabled robots provide massive help in defence as these can make decisions based on sensory feedback. Advances in vision systems, force and tactile sensors, speech recognition and even olfactory receptors enable technologies for collaborative robots, advanced mobile robotics and autonomous driving. Their force-limiting capability has led to the emergence of robotic systems that can safely work alongside humans. Machine vision technology, laser scanners, and imaging and mapping software support robots are making their way into defence applications.

Amit Maheshwari, robotics software engineer at Sastra Robotics, says that, LiDAR, or light detection and ranging, sensor is in demand because of its better tracking, imaging and mapping capabilities. In future we may have robots and drones helping our armies in every area. Computer machine learning and artificial intelligence (AI) in robots are the key emerging technologies.

Maheshwari adds, sensors are beneficial to mimic human cognitive recognition in robots. These help robots in interacting with the environment and humans by creating eye-sight vision and providing touch capabilities. The main challenge is with security while controlling over-Cloud platforms and the Internet.

LiDAR scanner sends out its own light to interact with the object and return to the sensor. It is not affected by environmental effects like colour or sunlight. The three major criteria that differentiate most laser scanners are range, angular resolution and speed.

LiDAR shines a laser off a rotating mirror. As the mirror spins, the laser scans a viewing angle between 70 and 360 degrees, effectively creating a fan of laser light around the sensor. Any object that comes within this range reflects the laser light back to the sensor. Distance is calculated based on how long the light takes to bounce back to the sensor. Speed of light is constant and known, so the instrument can calculate the distance between itself and the target with high accuracy.

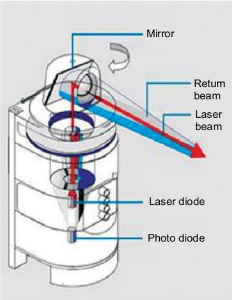

Fig. 4: LiDAR sensor demonstrating the time of flight principle (Credit: www.robotics.org)

Distance of the object = Speed of light × Time taken to bounce back/2

Types of sensors

Many basic sensors in robotics help the latter perform basic functions. These are:

1. A light sensor detects light and creates a voltage difference.
2. Sound is detected by a sound sensor by generating a voltage proportional to sound level.
3. Change in temperature provides voltage difference that is detected by a temperature sensor.
4. A contact sensor triggers the desired circuit on physical contact with any object.
5. A proximity sensor detects the presence of a nearby object within a given distance, without any physical contact, by analysing the return signal from interruptions.
6. A pressure sensor is sensitive to touch, force and pressure.
7. Approximate position of a robot is found by positioning sensors like Global Positioning System (GPS).
8. An accelerometer measures acceleration and tilt. Both static force and dynamic force affect an accelerometer.
9. A gyroscope measures and helps maintain orientation using the principle of angular momentum.

A new sensor technology, GelSight, has been developed by Massachusetts Institute of Technology (MIT) scientists. It is mounted on the gripper of robot arms to provide a detailed 3D map of the surface of the object in physical contact. It has given robots greater dexterity and sensitivity.

Sensors with military might. Some robots are primarily designed for explosive ordnance disposal (EOD) for military deployments. Tele-operated mobile bomb disposal robots utilise force sensors to enable operators to feel remote objects through end-of-arm force sensing and a haptic controller.

Applications of drones

Sensors embedded in an unmanned aerial vehicle (UAV) can be exposed to high accelerations and repetitive shocks, during take-off and landing, respectively. In some cases, the inertial measurement units (IMUs) are directly in contact with the outer air. A flying system is based on one crucial point: the lower its weight and power consumption, the longer it flies.

DJI Mavic air foldable drone was launched with amazing features such as 4K camera, smart gesture recognition, 3-axis gimbal, 3-dimensional obstacle detection, maximum flight time of 21 minutes with constant speed and 10km maximum flight range.

UAVs, or autonomous drones, perform tasks such as scouting, mapping, target tracking and offensive neutralising. These have the advantage of flying over medium (30 minutes) to long periods of time (up to 20-30 hours for larger units) without any need for a pilot inside. As a result, UAVs have three critical requirements: high stability during take off, navigation maneuvers and landing. These need integration of many high-performance sensors.

Autopilot integrates sensors corresponding to the three above-mentioned requirements. Main functional component of the autopilot is an